

Element E: Application of STEM Principles and Practices

Sponsor

- Mr.Eisenberg, one of our sponsors and our Engineering teacher, gave us help in the STEM principles and practices we should incorporate in our project through our group check-ins and conversations.

Balance - A main component of the bookend was to have good balance. This means to be stable even with several books leaning on it. To accomplish this, we put a base on the side where the books will stand. The weight of the books will hold down the plate attached to the bookend, simultaneously causing the bookend to stay stood up.

Structural Integrity - This component goes hand in hand with balance. The structure of the bookend greatly affects the stability of the bookend. To incorporate better structure, we decided to use a triangle for the base shape of the bookend. The triangular shape is known to bring the most balance to a structure, and is used in many structures such as bridges.

Weight Distribution - The bookend must be able to support several heavy books at a time. The weight distribution in the bookend is what we paid most attention to. The weight of the triangle base cannot be supported by itself. Because of this, as well as taking into reference known designs, we decided to incorporate a solid plate at the base to distribute the weight evenly compared to the triangular base.

Friction - An issue mentioned was the lack of friction in the base of the bookends, causing them to slide and move easily when books are placed on them. The bottom material that prevented this, such as a thin layer of foam, deteriorated over time. To prevent this from happening in our design, we decided to put four support pillars made of cork on each corner. Cork is significantly sturdier than foam, and we can also limit the material use by putting tall pillars on each corner instead of a thin layer covering the entire bottom surface.